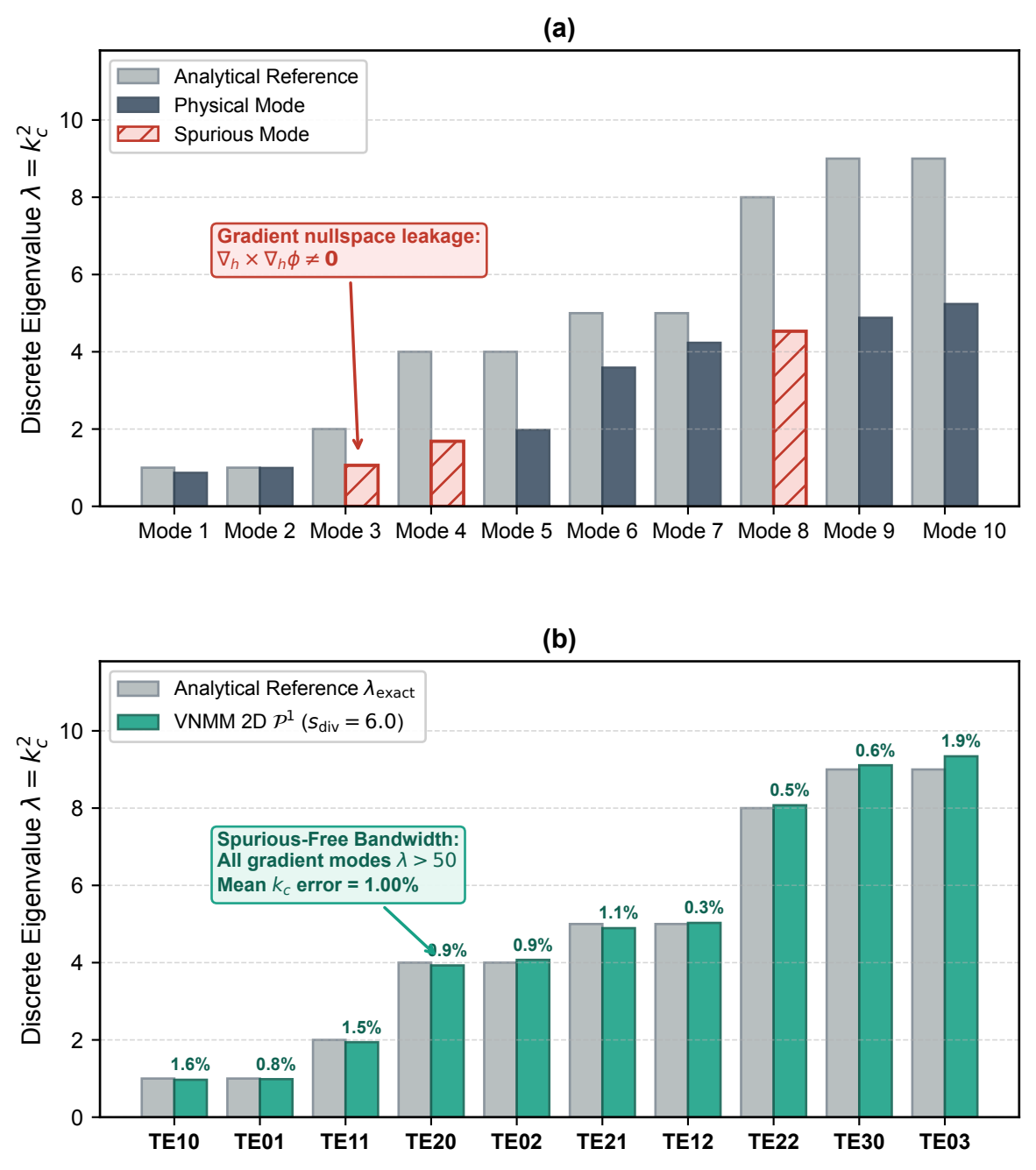

(a)
Analytical Reference
Physical Mode
Spurious Mode
Gradient nullspace leakage:
$\nabla_h \times \nabla_h \phi \neq \mathbf{0}$
Discrete Eigenvalue $\lambda = k_c^2$
0
2
4
6
8
10
Mode 1
Mode 2
Mode 3
Mode 4
Mode 5
Mode 6
Mode 7
Mode 8
Mode 9
Mode 10
(b)
Analytical Reference λ_{exact}
VNMM 2D $\mathcal{P}^1$ ($s_{\text{div}} = 6.0$)
Spurious-Free Bandwidth:
All gradient modes $\lambda > 50$
Mean k_c error = 1.00%
Discrete Eigenvalue $\lambda = k_c^2$
0
2
4
6
8
10
1.6%
0.8%
1.5%
0.9%
0.9%
1.1%
0.3%
0.5%
0.6%
1.9%
TE10
TE01
TE11
TE20
TE02
TE21
TE12
TE22
TE30
TE03